

Resource-Aware Keyword Spotting with MATLAB and MFCC Features

A MATLAB study of MFCC front ends, speaker-disjoint evaluation, and compact keyword-spotting models for resource-constrained audio systems.

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Project Overview

This project investigates a compact keyword-spotting system that recognises ten spoken commands - down, go, left, no, off, on, right, stop, up, and yes - plus unknown speech and silence. Keyword spotting is the small-vocabulary front end used by voice-controlled devices before a larger speech recogniser is activated. It must be accurate enough to avoid false activations, but computationally small enough to run continuously on resource-constrained hardware. The proposed solution is a reproducible MATLAB pipeline covering audio preparation, Mel-frequency cepstral coefficient (MFCC) extraction, baseline classification, and later comparison with a depthwise-separable convolutional neural network (DSCNN). The design is inspired by Texas Instruments' 12-class Google Speech Commands example for the MSPM0G5187, while the present project remains a PC-side simulation until separate embedded deployment has been implemented and tested.

Background and Motivation

Always-on keyword spotting illustrates an important acoustic engineering trade-off: stronger models can improve recognition, but they require more memory, computation, and energy. MFCCs provide a compact representation of the short-term spectral envelope on a perceptually motivated Mel scale. Davis and Mermelstein's early comparison established the usefulness of Mel-frequency cepstral representations for word recognition. Sainath and Parada showed that small-footprint CNNs can reduce false rejects relative to DNN baselines, while Tucker et al. demonstrated model compression without increasing runtime cost. More recently, Zhang et al. showed that depthwise-separable CNNs can achieve strong keyword-spotting accuracy under microcontroller resource constraints. This project was chosen because it combines core ELEC5305 concepts - framing, windowing, Fourier analysis, filterbanks, cepstral features, and objective evaluation - with a practical edge-audio application.

Proposed Methodology

The experiments will use Speech Commands v0.02, a public dataset containing 105,829 one-second spoken-word recordings at 16 kHz. The supplied official validation and testing lists will be respected so recordings from the same speaker do not leak across partitions. Ten commands will be retained as known classes. Other word folders will form the unknown class, and one-second regions from the background-noise recordings will form the silence class.

MATLAB will first standardise each clip to one second at 16 kHz. The acoustic front end will follow the TI example: 30 ms frames, 20 ms frame step, 40 Mel filters, and 10 MFCC coefficients. This produces 49 time frames by 10 coefficients for each utterance. A transparent k-nearest-neighbour classifier will provide a smoke-test baseline and verify the complete data path without requiring a deep-learning toolbox. The next model will be a compact DSCNN that operates directly on the 49 by 10 feature map. Controlled experiments will compare the baseline and DSCNN using identical data partitions.

Evaluation will report overall test accuracy, per-class precision and recall, macro recall, and a confusion matrix. Special attention will be given to the unknown and silence classes because they determine false activations in a realistic always-on system. Runtime, parameter count, and estimated memory will be recorded for resource awareness. Repeated training seeds will be used before drawing model-ranking conclusions. Hardware compilation or energy measurements will be reported only if they are later performed on the named target under a separate, documented protocol.

Expected Outcomes

The project will deliver a runnable MATLAB prototype, a traceable subset manifest, MFCC and spectrogram visualisations, reproducible evaluation metrics, and a GitHub Pages site explaining the design and results. The minimum outcome is a verified end-to-end baseline that performs above the 12-class chance level while maintaining speaker-disjoint evaluation. The intended final outcome is a DSCNN that improves accuracy and macro recall over the classical baseline with an explicitly reported computation and memory cost. A final comparison will identify which errors arise from acoustically similar commands and whether the unknown class needs stronger augmentation or decision thresholds.

Timeline (Weeks 1-13)

Weeks	Work package
1-2	Define the 12-class task, audit the dataset, and review literature.
3-5	Implement and verify audio loading, official splits, MFCC extraction, and the baseline.
6-9	Implement the DSCNN, establish repeated-seed training, and tune only on validation data.
10-11	Evaluate test performance, runtime, memory, robustness, and error patterns.
12-13	Finalise the report, reproducibility package, GitHub documentation, and demonstration.

References

1. P. Warden, "Speech Commands: A Dataset for Limited-Vocabulary Speech Recognition," arXiv:1804.03209, 2018.
2. Y. Zhang, N. Suda, L. Lai, and V. Chandra, "Hello Edge: Keyword Spotting on Microcontrollers," arXiv:1711.07128, 2017.
3. S. B. Davis and P. Mermelstein, "Comparison of Parametric Representations for Monosyllabic Word Recognition in Continuously Spoken Sentences," IEEE Transactions on Acoustics, Speech, and Signal Processing, vol. 28, no. 4, pp. 357-366, 1980, doi:10.1109/TASSP.1980.1163420.
4. T. N. Sainath and C. Parada, "Convolutional Neural Networks for Small-footprint Keyword Spotting," Proceedings of Interspeech, pp. 1478-1482, 2015, doi:10.21437/Interspeech.2015-147.
5. G. Tucker, M. Wu, M. Sun, S. Panchapagesan, G. Fu, and S. Vitaladevuni, "Model Compression Applied to Small-Footprint Keyword Spotting," Proceedings of Interspeech, pp. 1878-1882, 2016, doi:10.21437/Interspeech.2016-1393.
6. Texas Instruments, "Google Speech Command Recognition," Tiny ML Tensorlab 1.4.0 User Guide, 2026.

Preliminary Evidence

The first smoke test used 720 training and 240 held-out test utterances with a manual 5-nearest-neighbour classifier. It produced the required 49 by 10 MFCC map, found zero speaker overlap, and reached 39.17% accuracy and macro recall, above the 8.33% chance level. Silence recall was 100%, but unknown-word recall was 0%, showing that a simple closed-set distance classifier does not provide a reliable rejection boundary. This result verifies the pipeline while motivating the planned temporal model, data augmentation, and explicit unknown threshold; it is not presented as DSCNN or embedded-device performance.

Appendix: Preliminary Real-Data Results

The following figures were generated by the submitted MATLAB smoke test from held-out Speech Commands v0.02 samples. The result is a 5-NN baseline, not a DSCNN or embedded-device measurement.

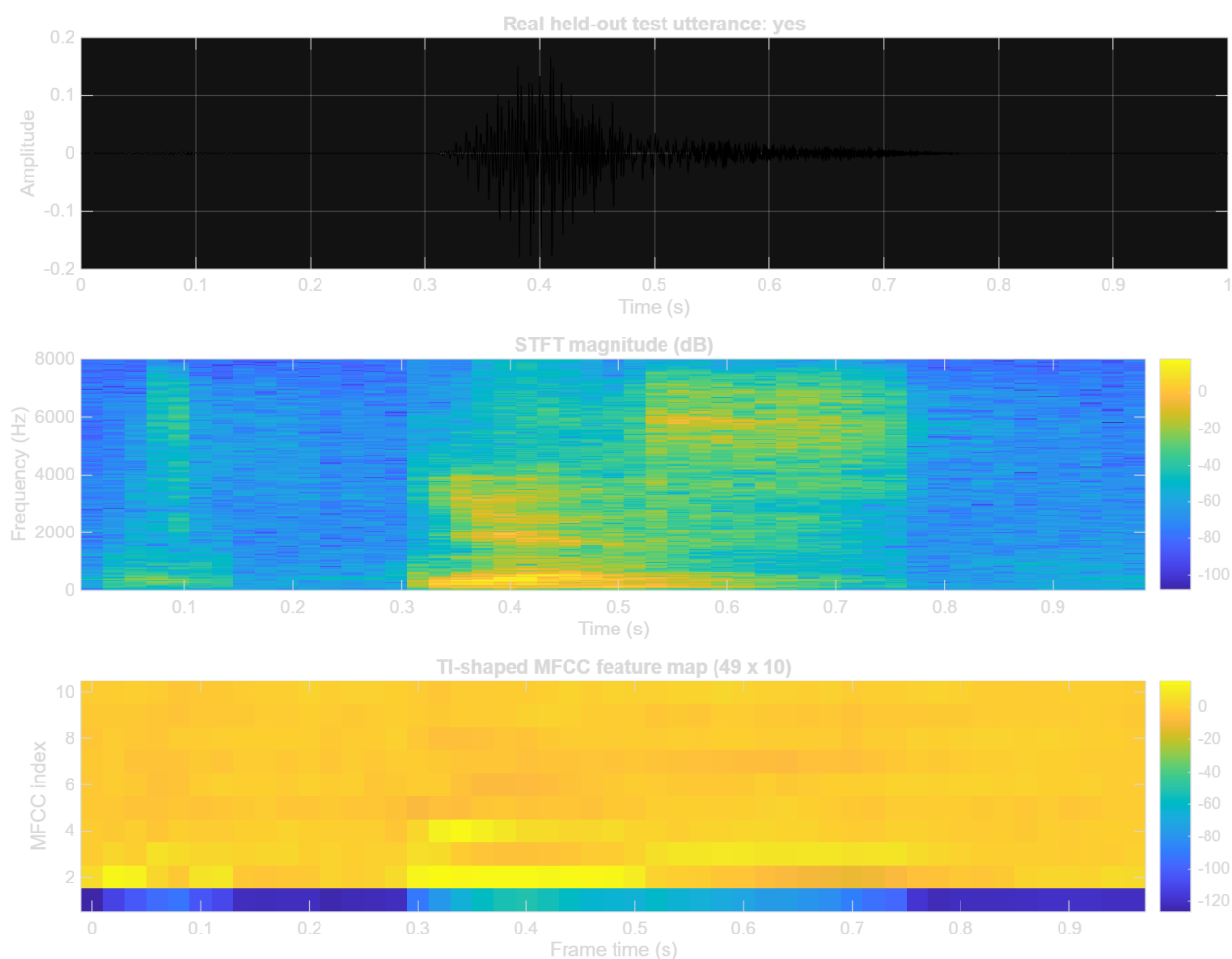


Figure A1. Real waveform, STFT, and TI-shaped 49 x 10 MFCC map.

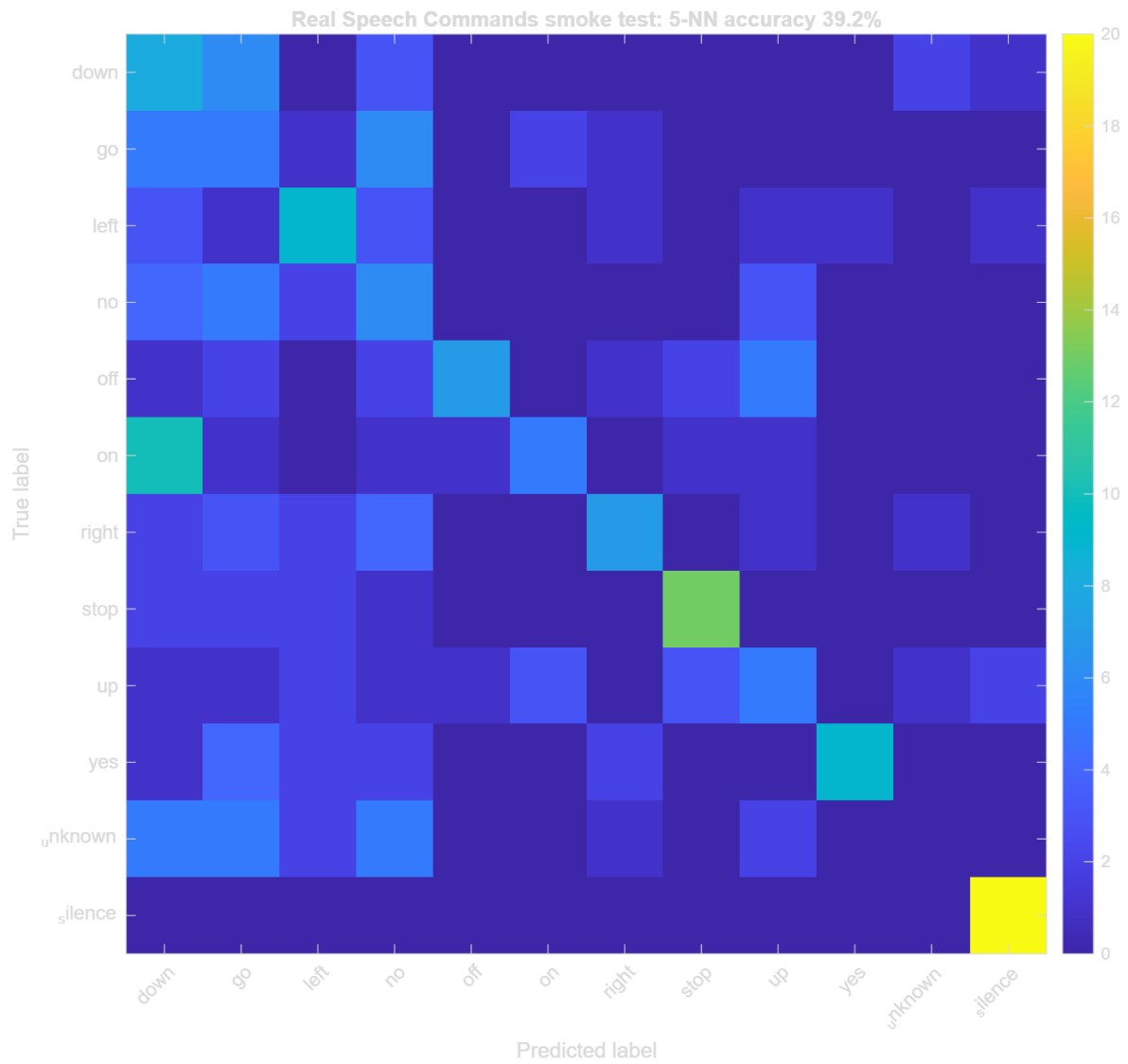


Figure A2. Speaker-disjoint 12-class smoke-test confusion matrix (240 test samples).